

■ 3.5.9 Defining Integrals

You can supplement the built-in integration functions of *Mathematica* by defining transformation rules for other integrals. You can, for example, enter some of the long lists of particular definite integrals that can be done.

The lack of a definite standard form for all integrals means that you sometimes have to give several different versions of the same integral: that is no guarantee that the appropriate changes of variable will automatically be done.

This integral cannot be done by the built-in *Mathematica* integration algorithm.

```
In[1]:= Integrate[Sin[Sin[x]], x]
Out[1]= Integrate[Sin[Sin[x]], x]
```

Before you add your own rules for integration, you have to remove write protection.

```
In[2]:= Unprotect[Integrate]
Out[2]= {Integrate}
```

You can, however, set up your own rule to define the integral to be, say, a “Jones” function.

```
In[3]:= Integrate[Sin[Sin[a_. + b_. x]], x] :=
        Jones[a, x]/b
```

Now *Mathematica* can do integrals that give Jones functions.

```
In[4]:= Integrate[Sin[Sin[3x]], x]
Out[4]=  $\frac{\text{Jones}[0, x]}{3}$ 
```